



M3 Screw



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M3 Screw



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M3 Screw



D



M3 Screw



D

Power Supply

Power Supply.SchDoc

Ethernet_PHY

Ethernet_PHY.SchDoc

SD

SD.SchDoc

CAN

CAN.SchDoc

Info_LEDs

Info_LEDs.SchDoc

MCU

MCU.SchDoc

Electrovalvula

Electrovalvula.SchDoc

Bomba_Refri

Bomba_Refri.SchDoc

Caudalimetro

Caudalimetro.SchDoc

Transductores_Presion

Transductores_Presion.SchDoc

Regulador_Presion

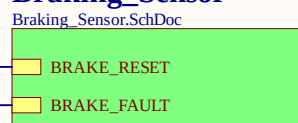
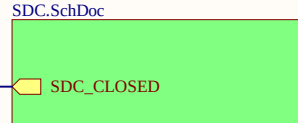
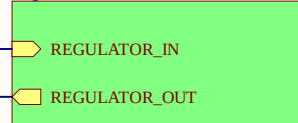
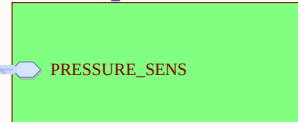
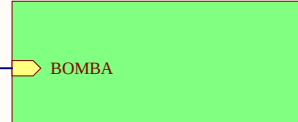
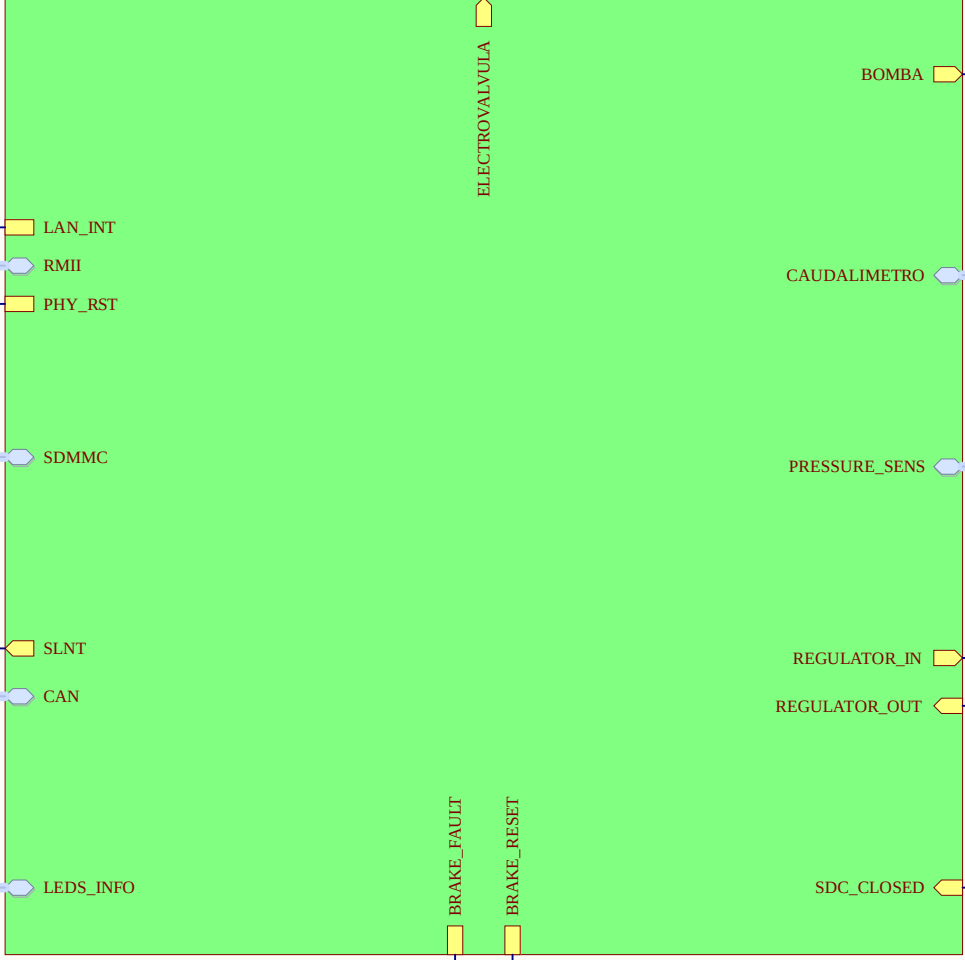
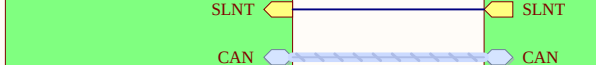
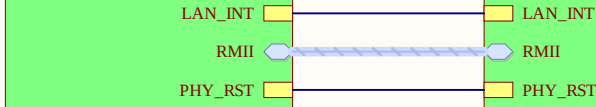
Regulador_Presion.SchDoc

SDC

SDC.SchDoc

Braking_Sensor

Braking_Sensor.SchDoc



PCB Manufacturing Notes

This PCB contains high speed analog and digital signals that should be routed with specific impedance profiles, using coplanar microstrip, stripline and differential routing techniques.

Details such as the specific impedance profile, trace width and trace spacing can be found in the layout notes in each sub-sheet of this PCB.

The impedance profiles are included in the Layer Stack Manager of this project and will only be accurate if the following aspects are specified when ordering:

Layers = 6
Inner copper weight = 0.5oz
Outer copper weight = 0.5oz + plating
PCB thickness = 1.6mm
Manufacturer = NCAB
Stack-up Code = 6ML BOARD - 1,6MM

You can find the details of the stackup 6ML BOARD - 1,6MM:

Contents	Thickness (mm)
Solder mask	0,02
0,5oz + plating	0,035
PP	0,200
CCL 0,43 1oz/1oz	0,035
	0,360
	0,035
PP	0,240
CCL 0,43 1oz/1oz	0,035
	0,360
	0,035
PP	0,200
0,5oz + plating	0,035
Solder mask	0,02

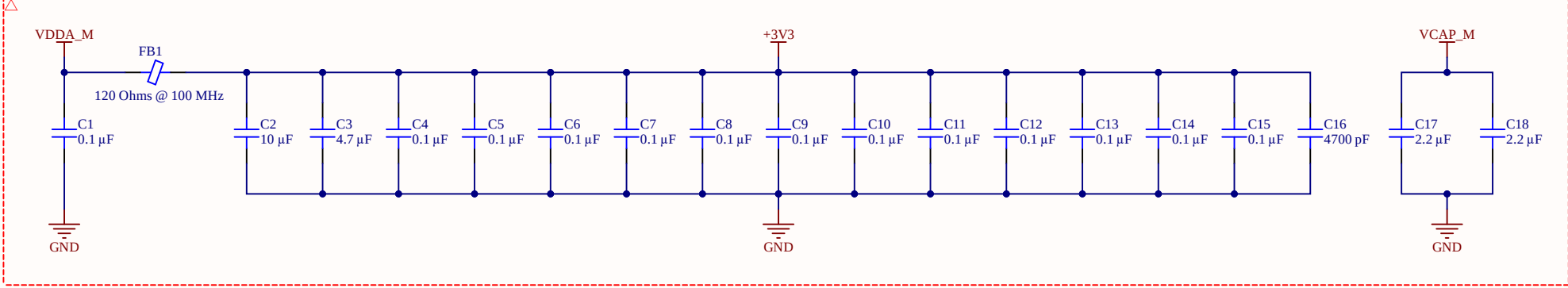
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Last Modification: 1/4/2026	Sheet 1 of 14	
Author: *		
Checked by: *		



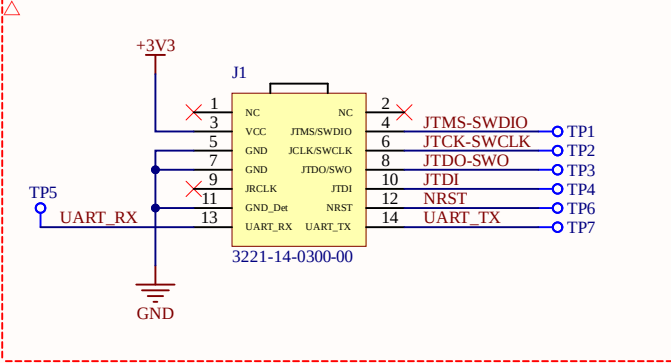
HYPERLOOP UPV



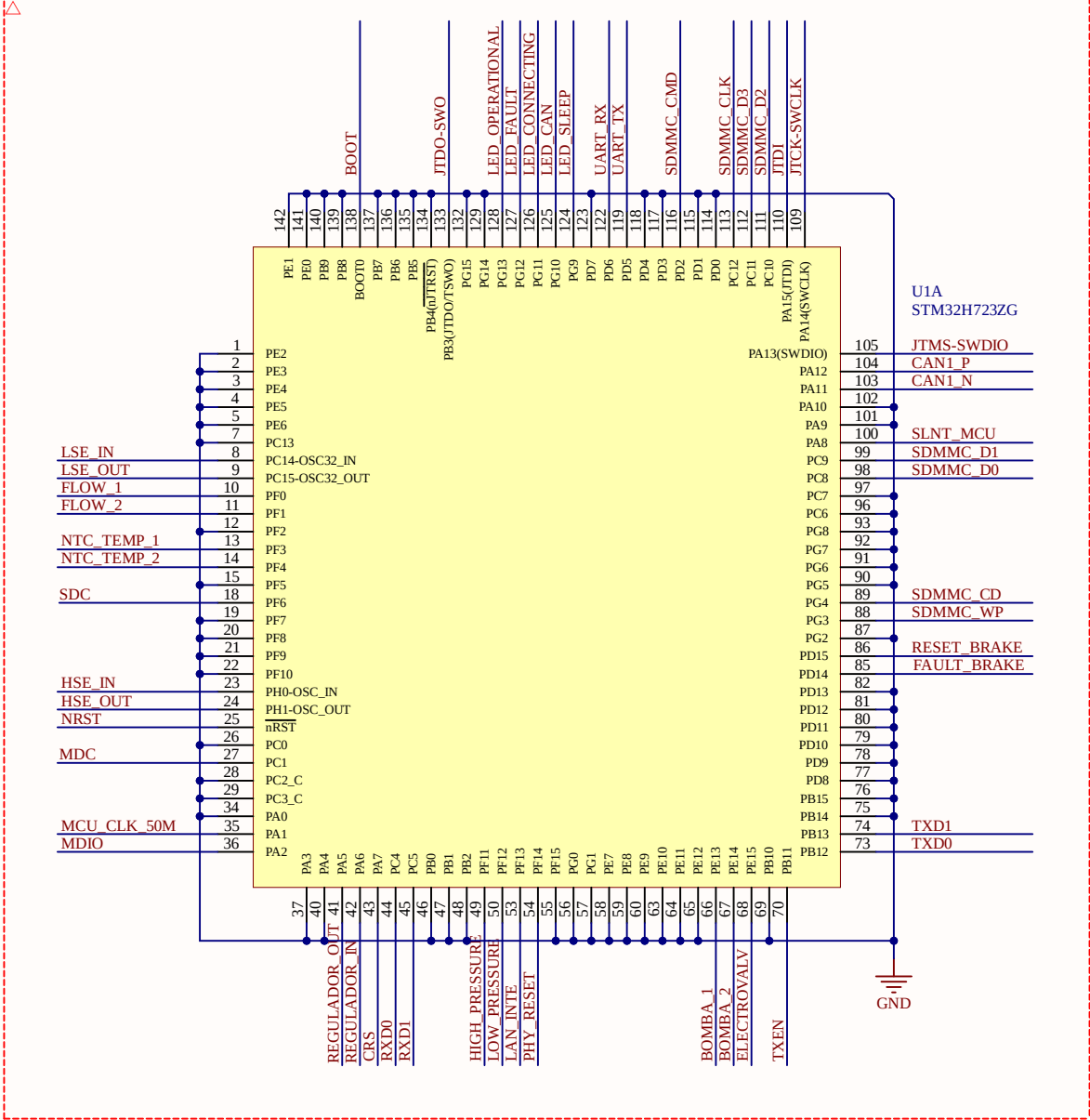
MCU Supply Decoupling



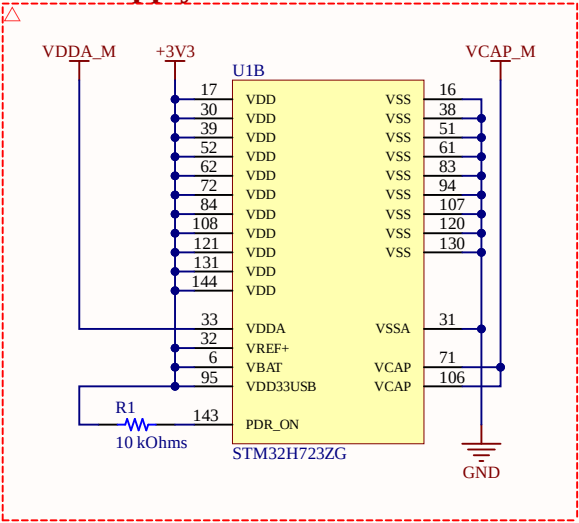
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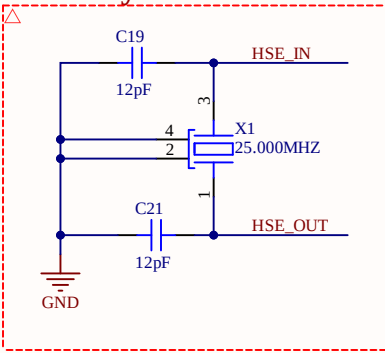
MCU



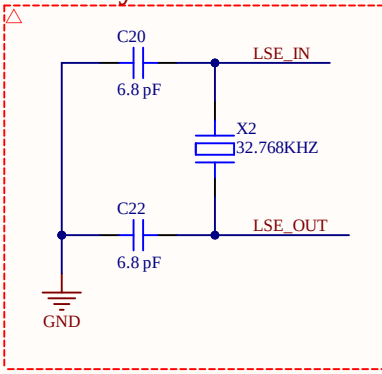
MCU Supply



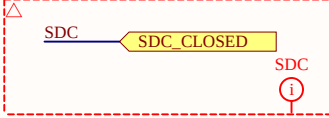
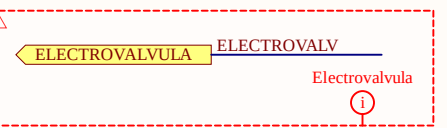
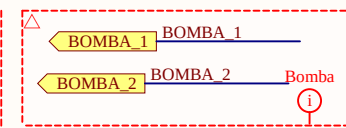
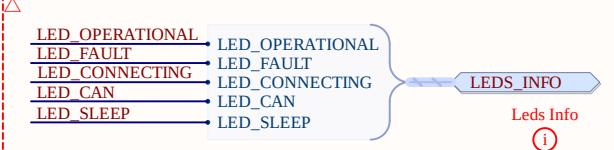
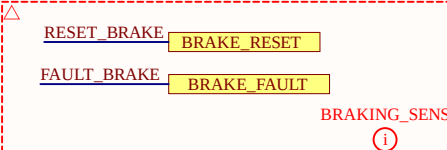
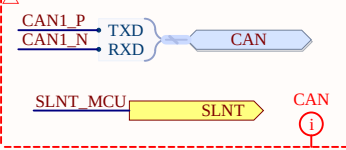
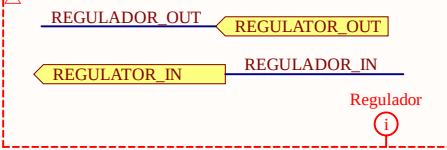
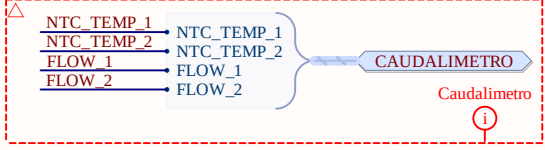
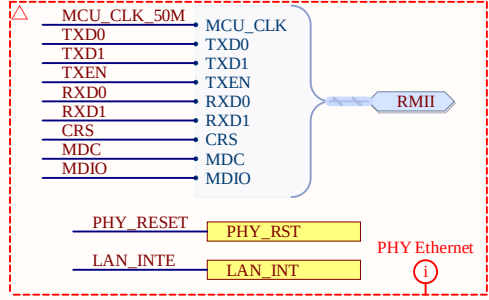
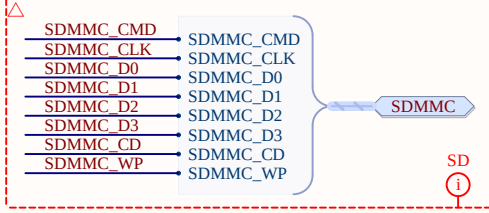
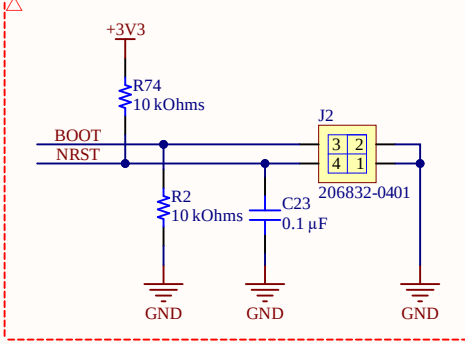
HSE Crystal



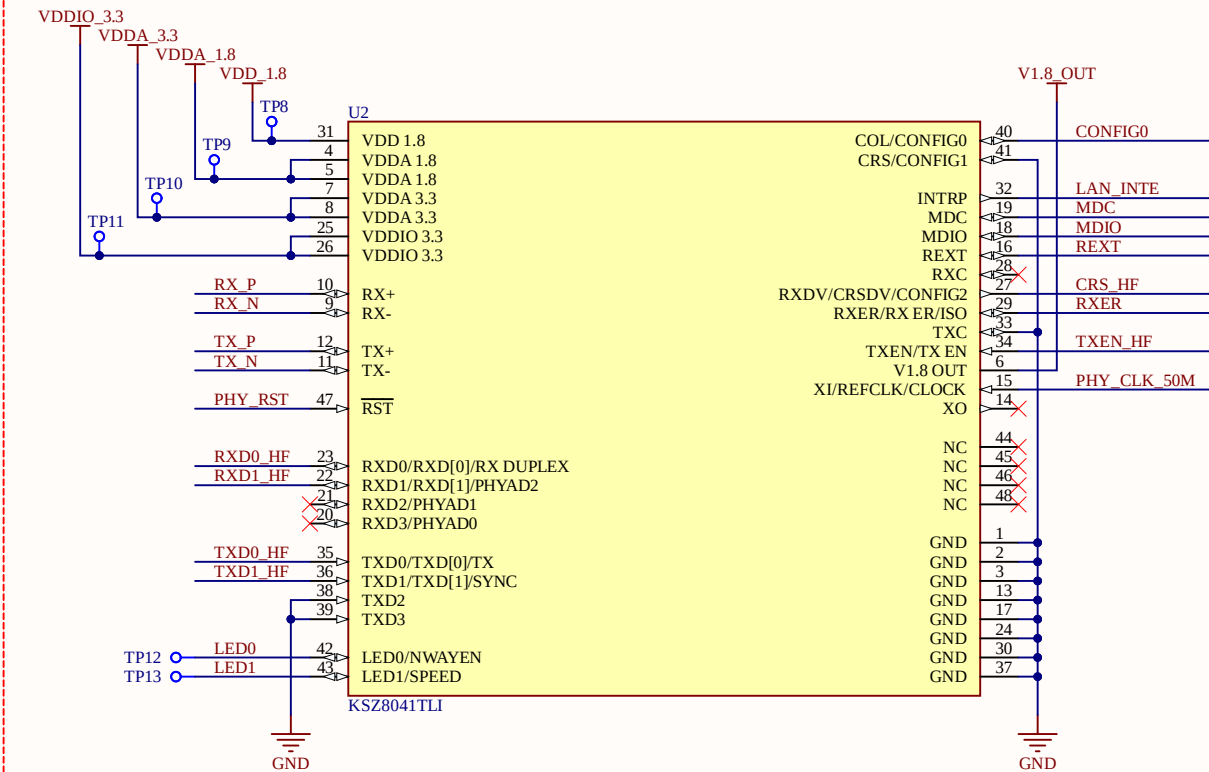
LSE Crystal



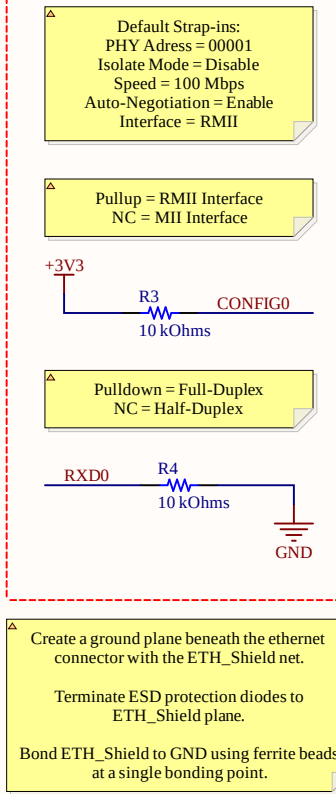
Boot



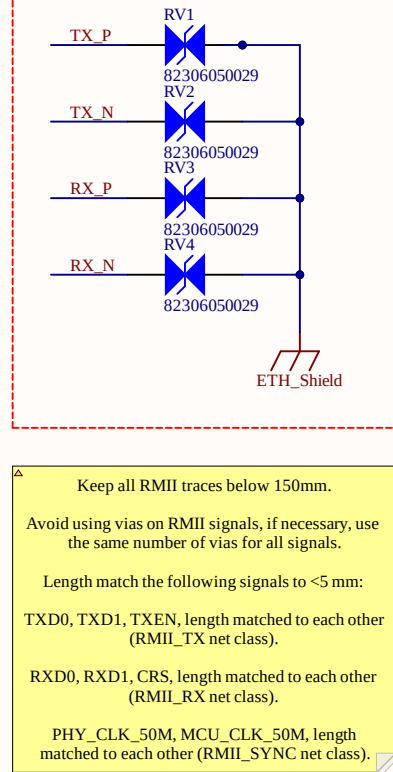
Ethernet Transceiver



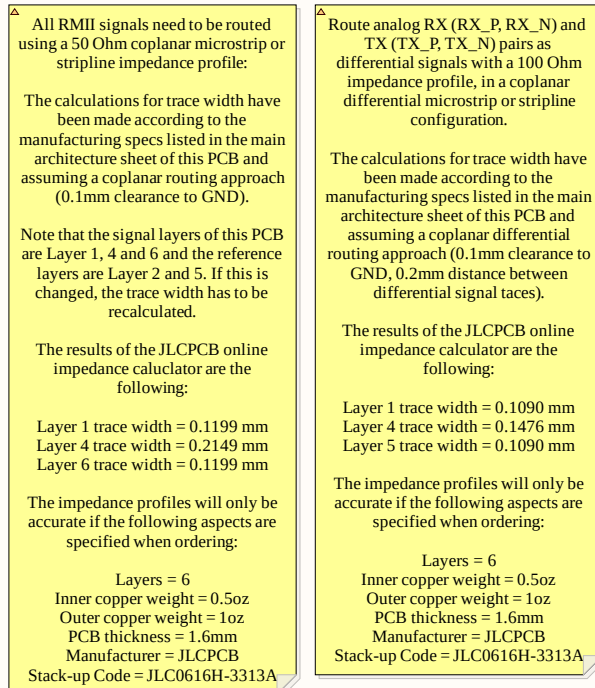
Strap-In Options



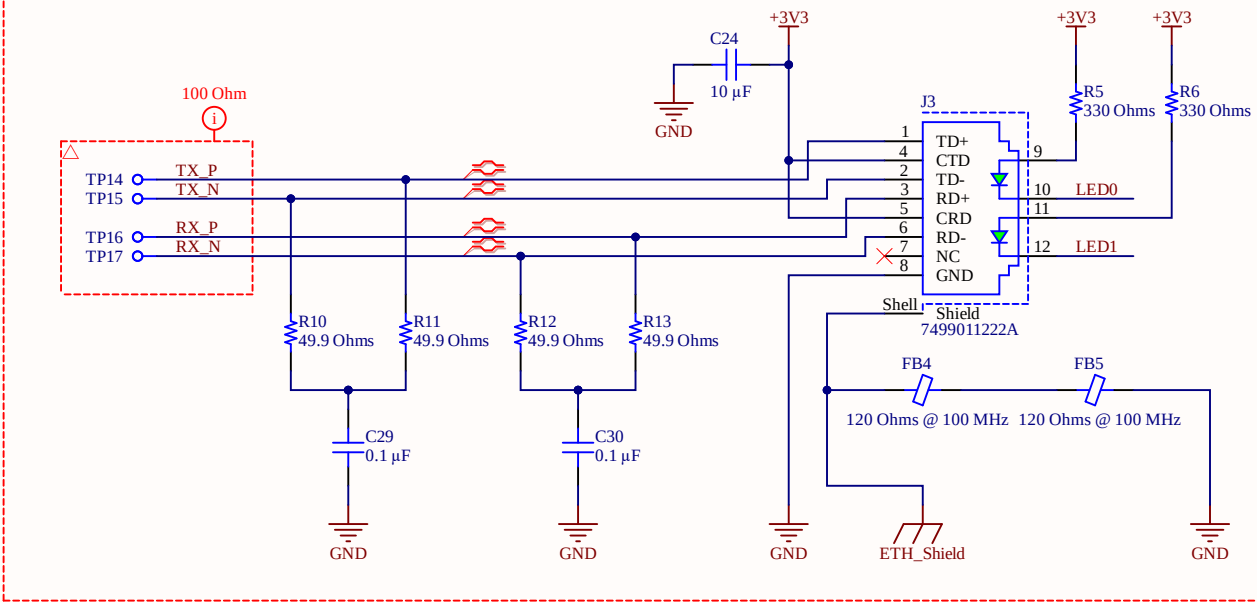
ESD Protection



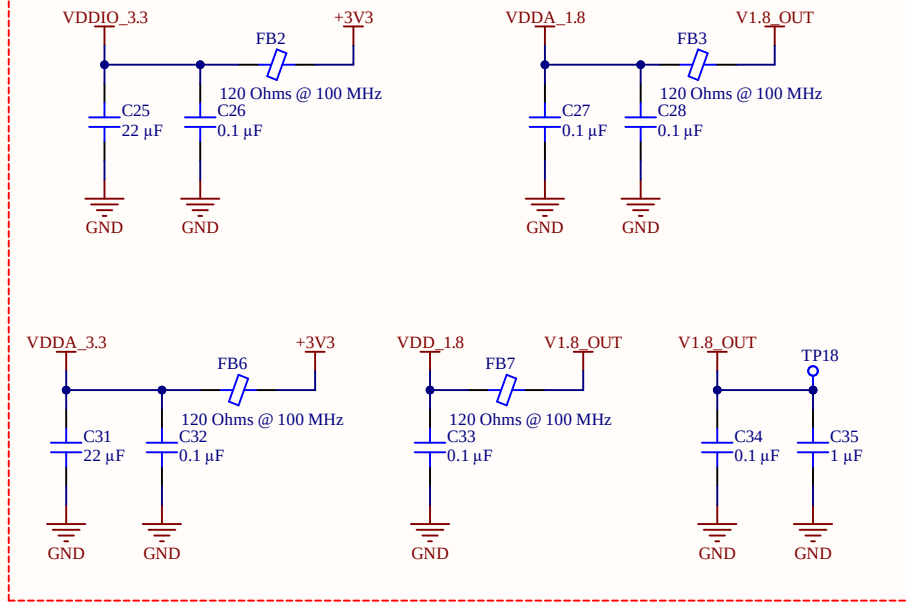
Layout Notes



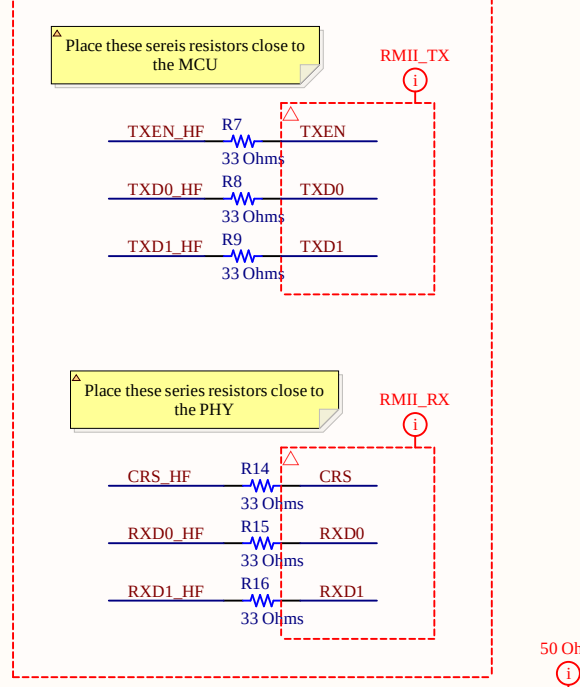
Analog Interface



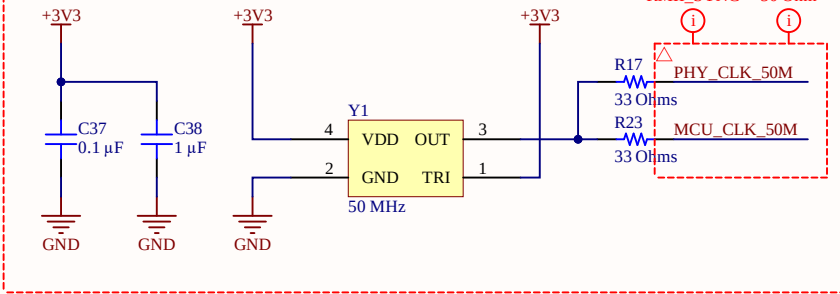
Transceiver Supply Decoupling



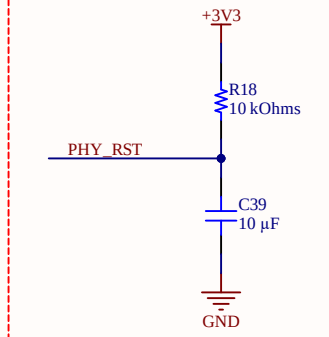
Slew Rate Control



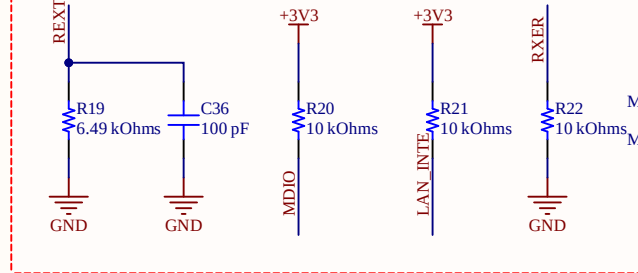
50 MHz Clock



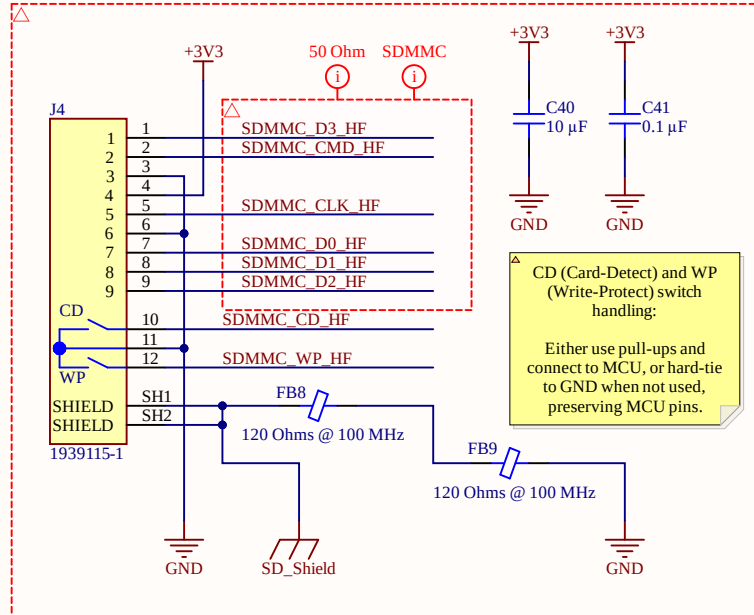
Reset



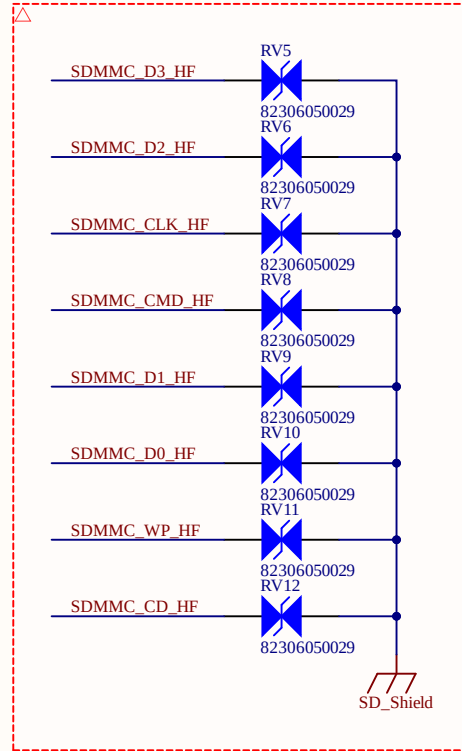
Biasing



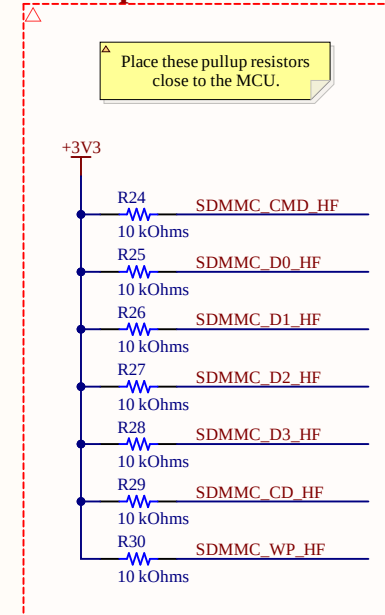
SD Card Slot



ESD Protection



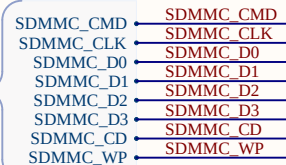
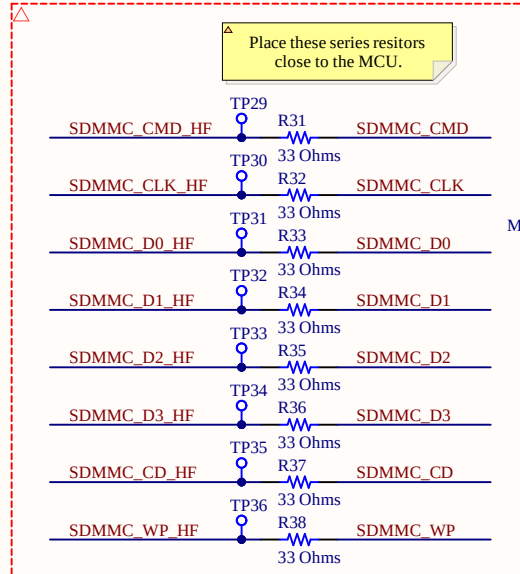
Pull-Up



Layout Notes

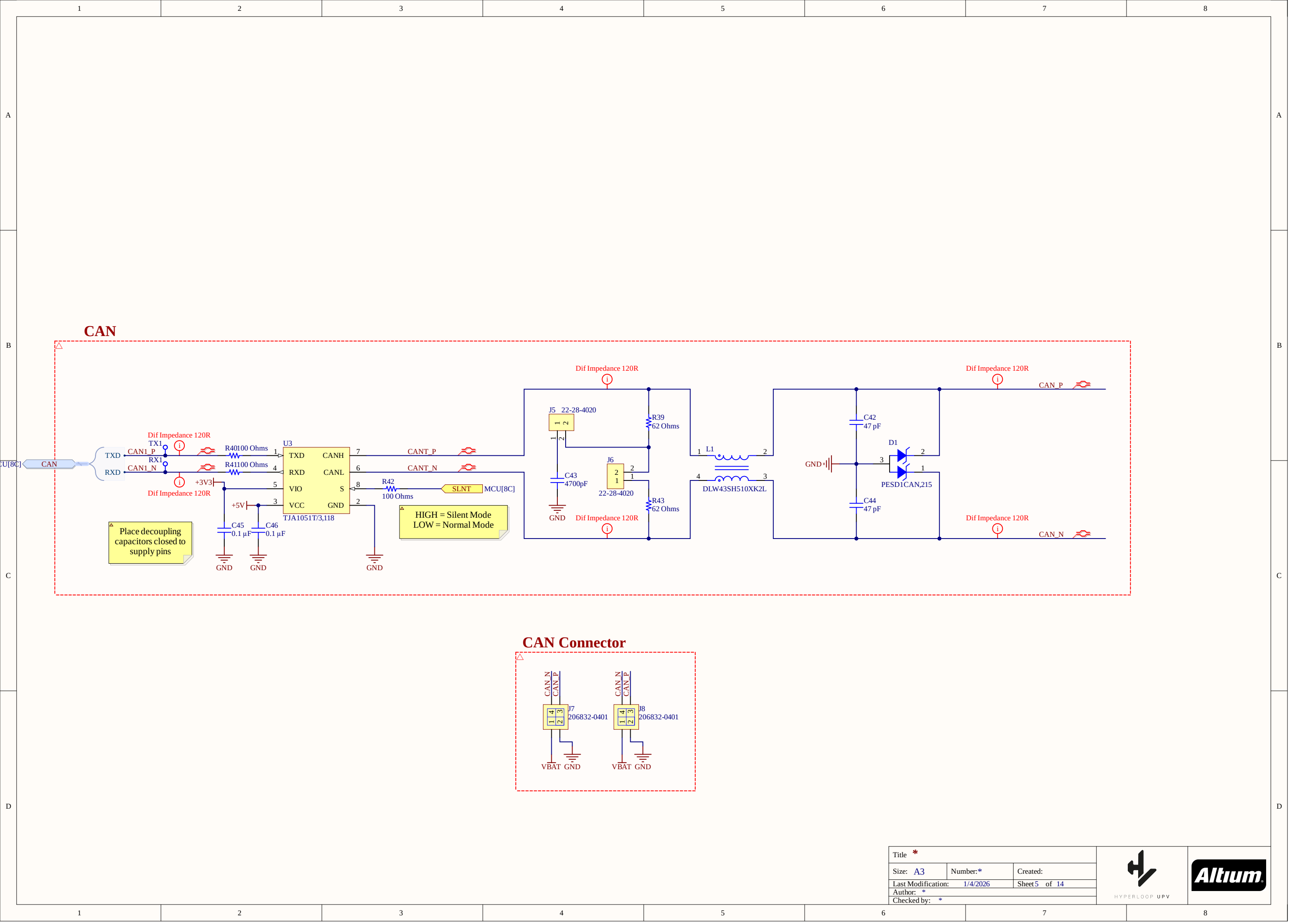
- Keep SDMMC traces below 120mm.
- Avoid using vias on SDMMC signals, if necessary, use the same number of vias for all signals.
- Length match the following signals to <10 mm:
CMD, CLK, D0, D1, D2, D3, length matched to each other (SDMMC net class).
- Create a ground plane beneath the SD card connector with the SD_Shield net.
- Terminate ESD protection diodes to SD_Shield plane.
- Bond SD_Shield to GND using ferrite beads at a single bonding point.
- The following signals need to be routed using a 50 Ohm coplanar microstrip or stripline impedance profile:
CMD, CLK, D0, D1, D2, D3.
- The calculations for trace width have been made according to the manufacturing specs listed in the main architecture sheet of this PCB and assuming a coplanar routing approach (0.1mm clearance to GND).
- Note that the signal layers of this PCB are Layer 1, 4 and 6 and the reference layers are Layer 2 and 5. If this is changed, the trace width has to be recalculated.
- The results of the JLCPCB online impedance calculator are the following:
Layer 1 trace width = 0.1199 mm
Layer 4 trace width = 0.2149 mm
Layer 6 trace width = 0.1199 mm
- The impedance profiles will only be accurate if the following aspects are specified when ordering:
Layers = 6
Inner copper weight = 0.5oz
Outer copper weight = 1oz
PCB thickness = 1.6mm
Manufacturer = JLCPCB
Stack-up Code = JLC0616H-3313A

Slew Rate Control

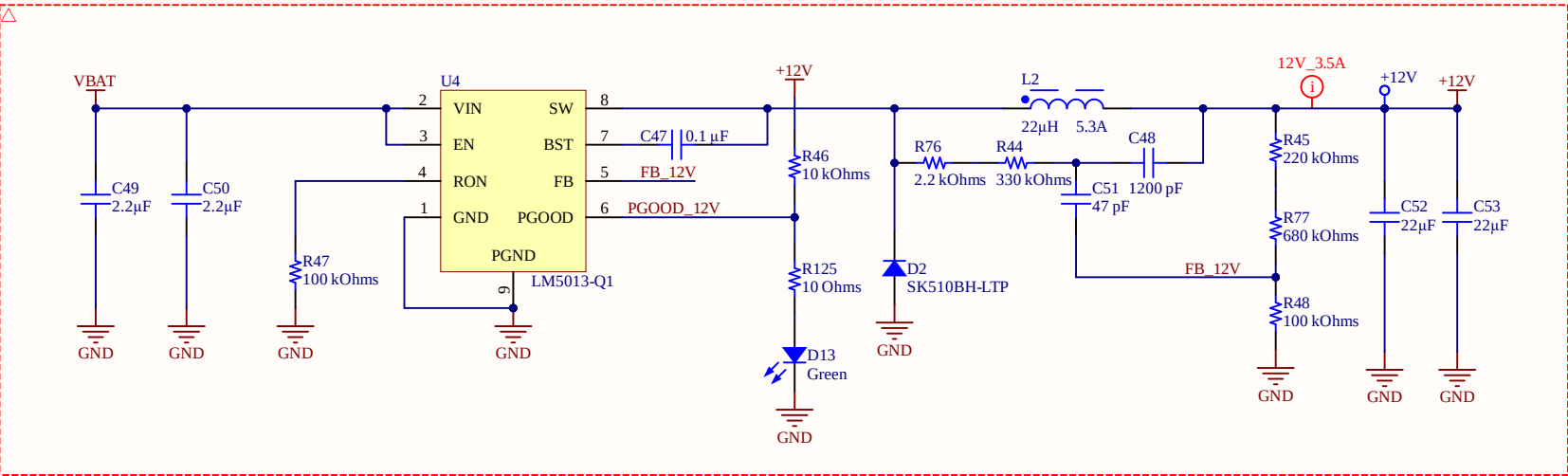


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Author: *		
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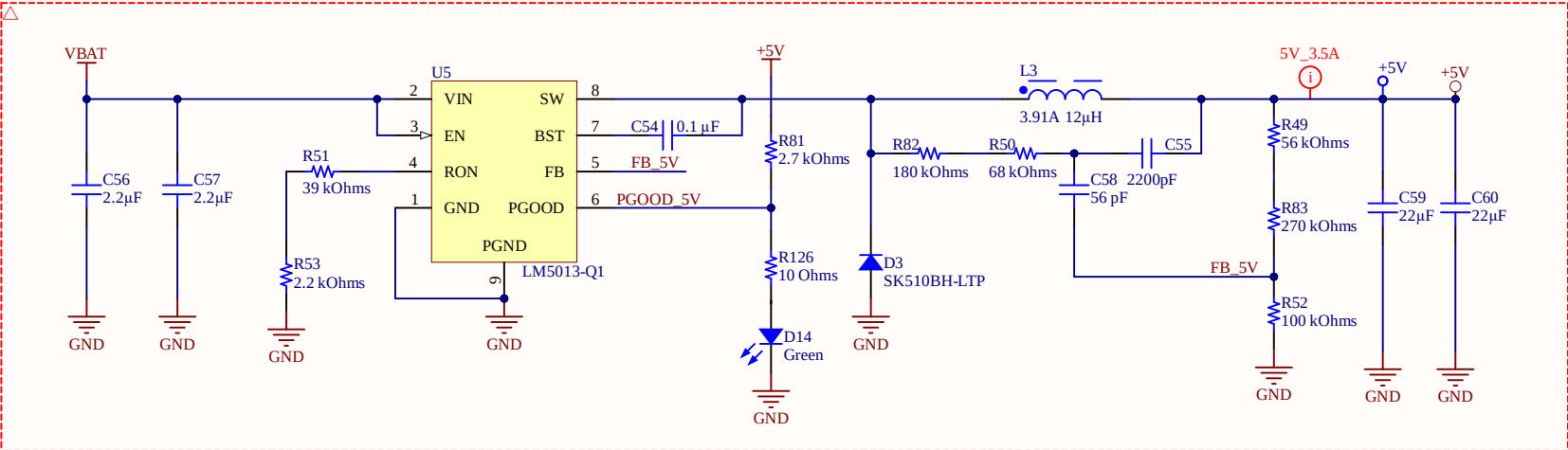




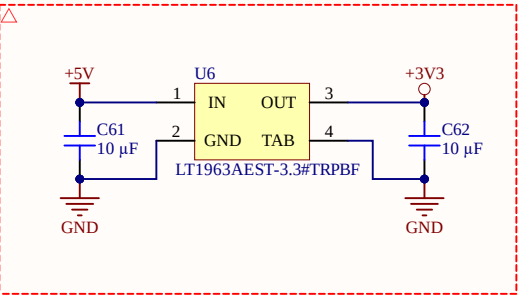
+12V Power Supply



+5V Power Supply



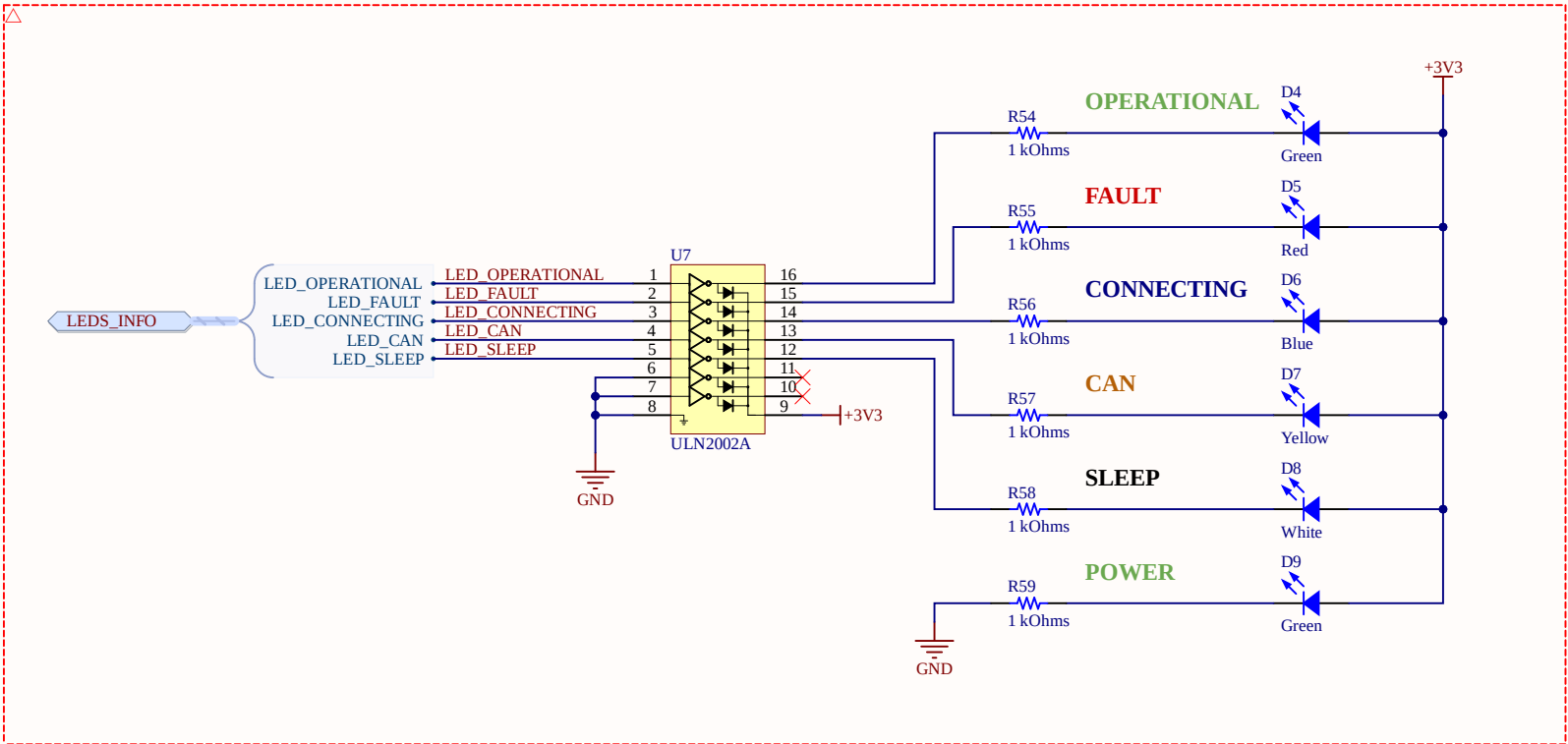
+3V3 Power Supply



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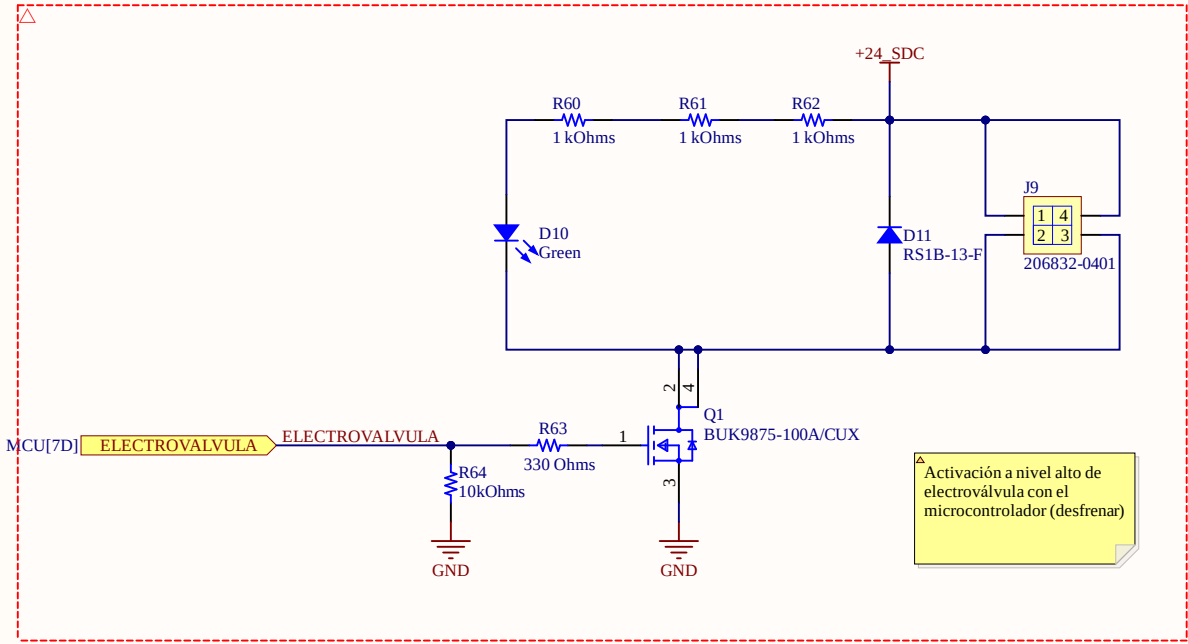
Information LEDs



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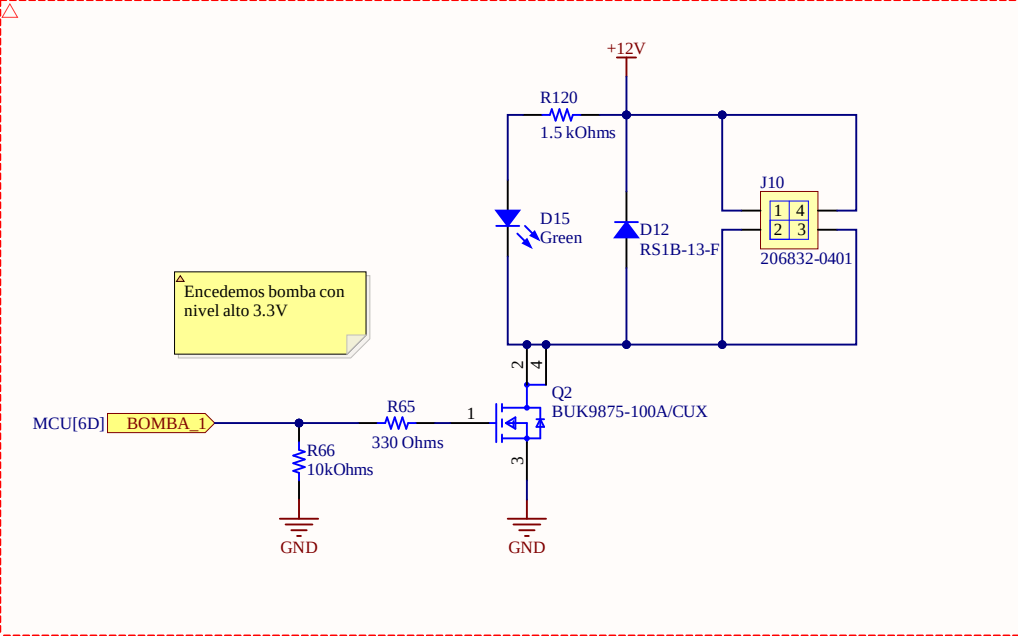
ELECTROVALVULA



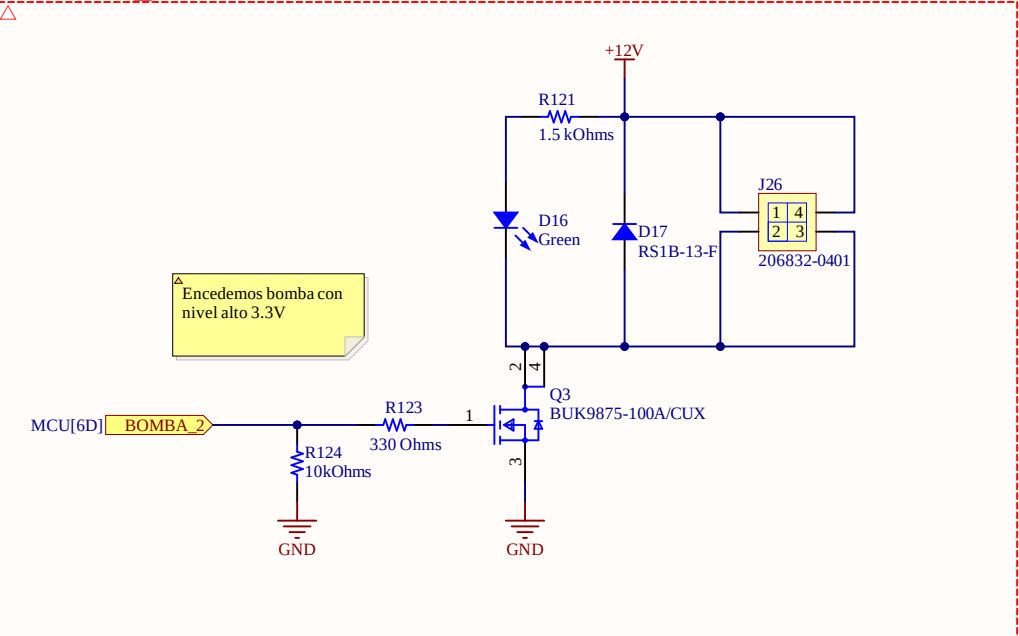
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BOMBA_1



BOMBA_2

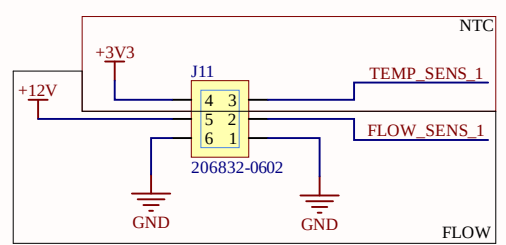


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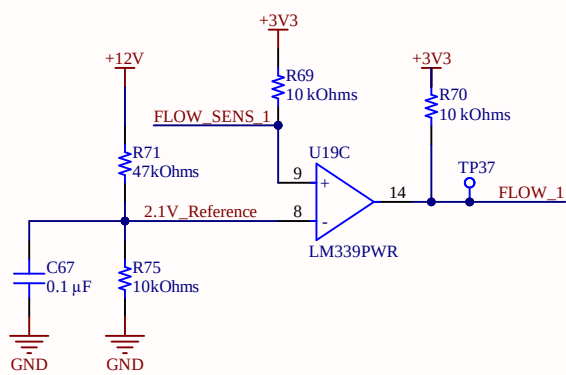
Caudalimetro_1

Conector_1

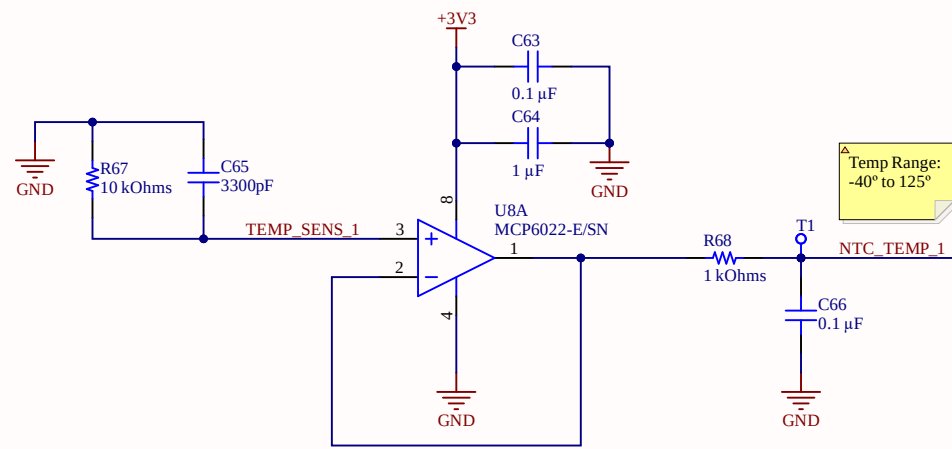


Como no sabemos si la salida digital del sensor de caudal es open drain o push pull, no nos podemos arriesgar a dañar el micro, por eso usamos este comparador, la idea es cuando se sepa en caso de que sea push pull no soldaremos la pull-up que se encuentra en el pin 5 (entrada positiva) y en caso de ser open drain soldaremos el comparador junto a la pull-up

Sensor_Caudal_1



Sensor_NTC_1

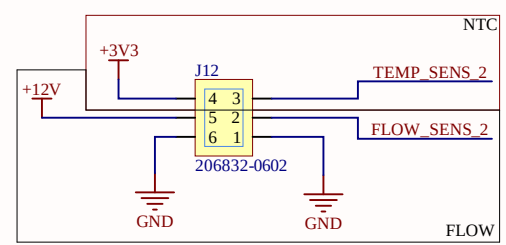


Adjunto aqui el link de la pagina web, en ella encontramos el datasheet y el manual: <https://shop.alphacool.com/en/shop/controllers-and-sensors/flow-sensor/17558-alphacool-es-flow-and-temperature-indicator-highflow-with-rpm-signal>

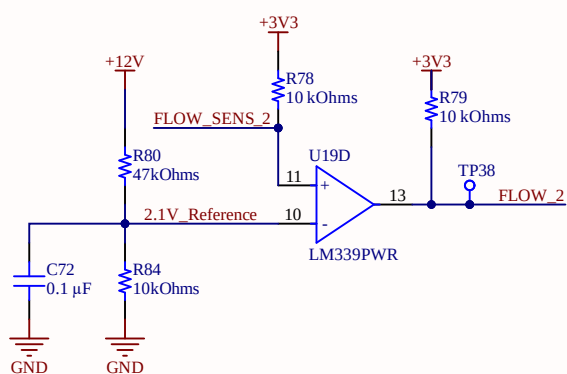
Explicando un poco este circuito del caudalimetro tenemos por un lado, un sensor de temp, el cual es una NTC que lee la temperatura del agua de la refrigeracion, y por otro lado tenemos un sensor de caudal, el cual es un sensor Hall con Tachometer/RPM Signal, esto indica que es salida digital Open Drain (por eso la pull-up), ademas este salida del sensor la deberemos de conector a un INPUT CAPTURE de la STM32, debido a que el codigo de la STM32 no mediremos voltaje, si no que el micro contará los pulsos que le llegan en periodo determinado para calcular la frecuencia (RPM)

Caudalimetro_2

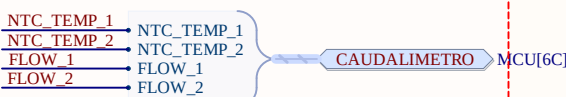
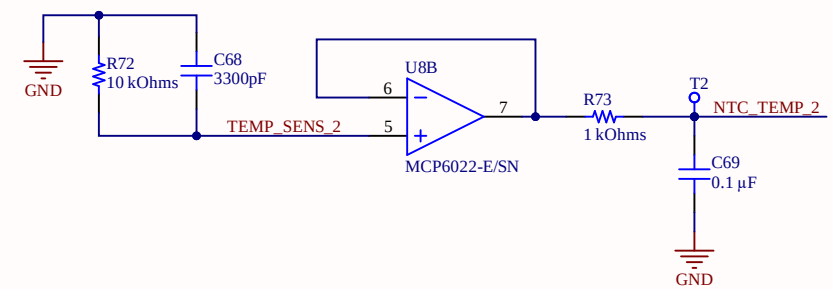
Conector_2



Sensor_Caudal_2

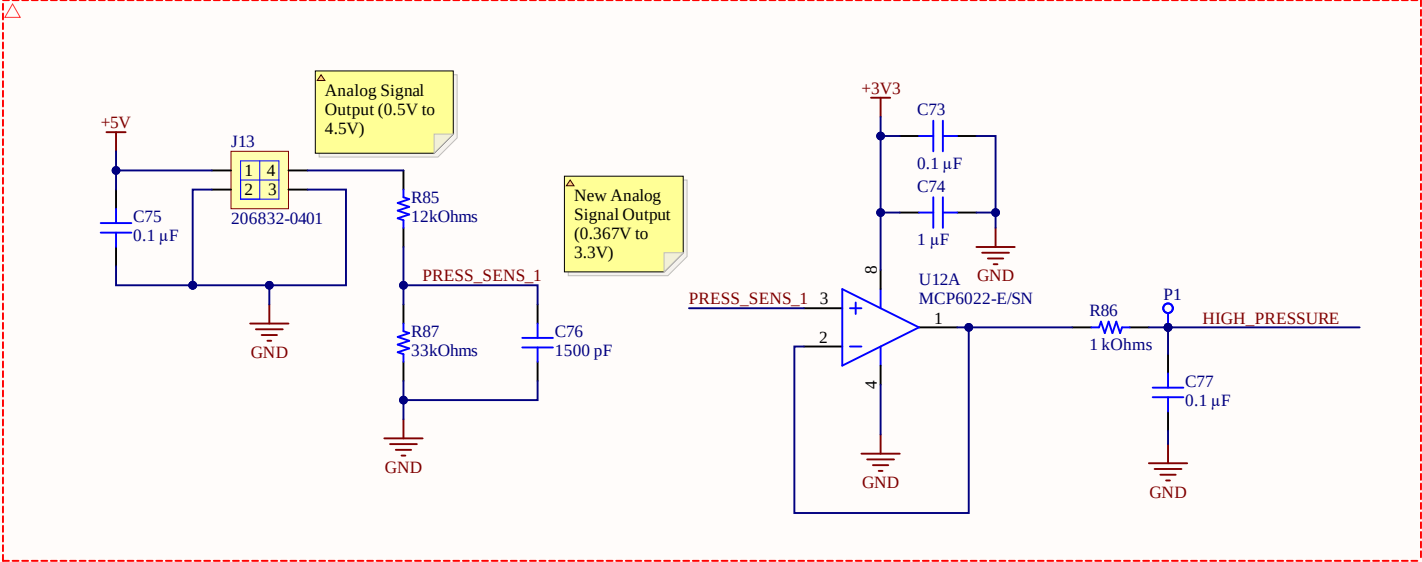


Sensor_NTC_2



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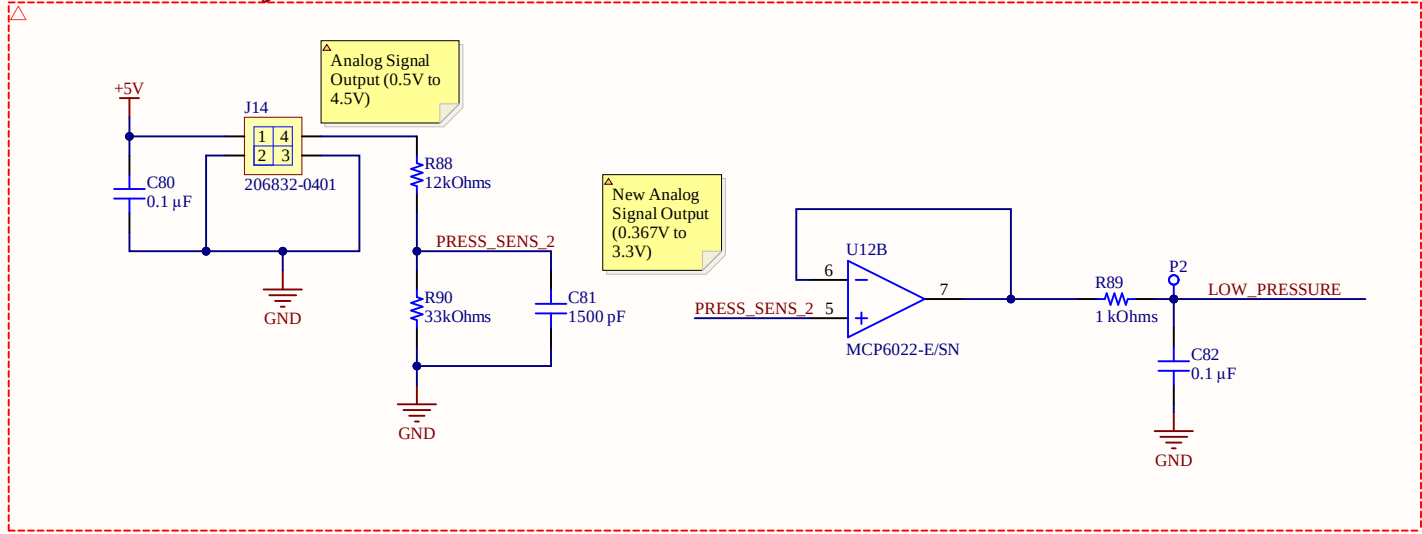
Transductor_Alta_Presion



Pin Out: Red = Vdc (5V) 10% to 90% Vs
Black = GND
Green =No se usa
White = Analog Signal Output (0.5V to 4.5V)

Ratiometric
Rango de presion (0-5000 Psi)

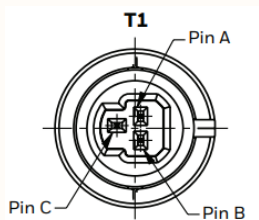
Transductor_Baja_Presion



Pin Out:
Pin A = Vdc (5V) 10% to 90% Vs
Pin B = GND
Pin C = Analog Signal Output (0.5V to 4.5V)

Ratiometric
Rango de presion (0-10 Bar)

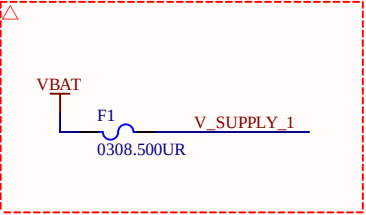
Como es Sealed Gage (parte de 1 atm = 1.013 Bar) no parte realmente de 0 a 10 Bar, si no que el rango de presion real (1.013-11.013 Bar)



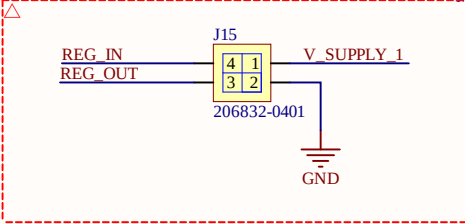
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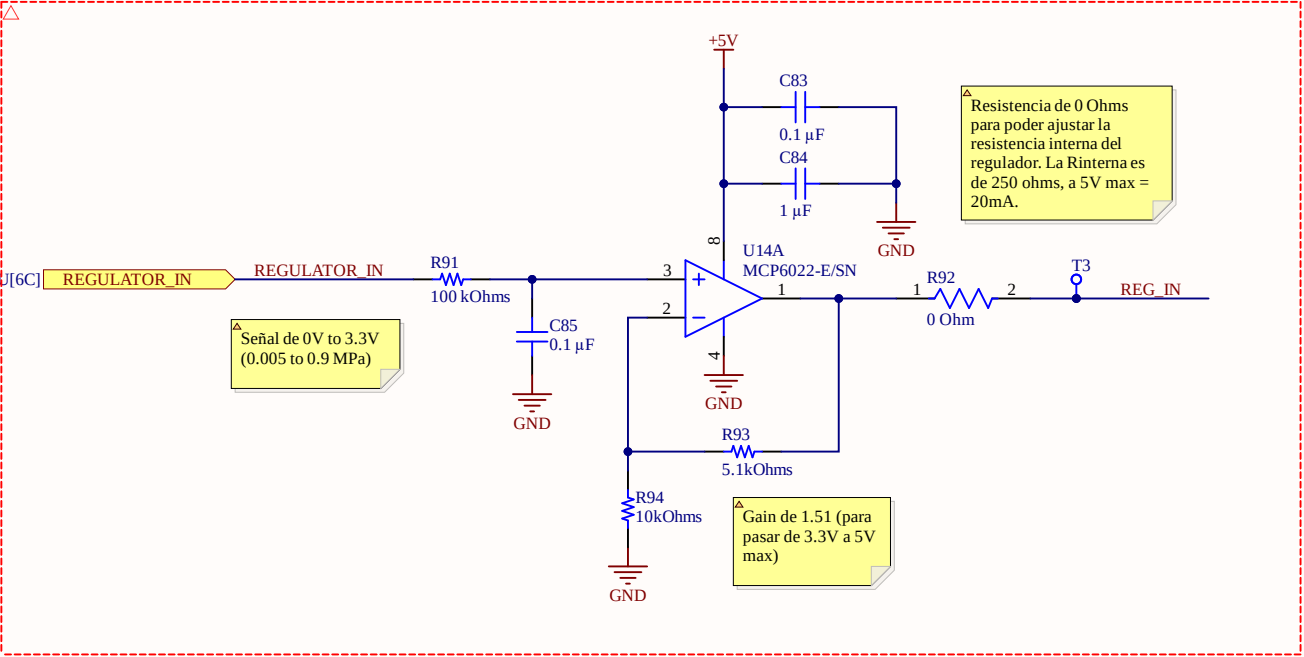
Supply



Conector



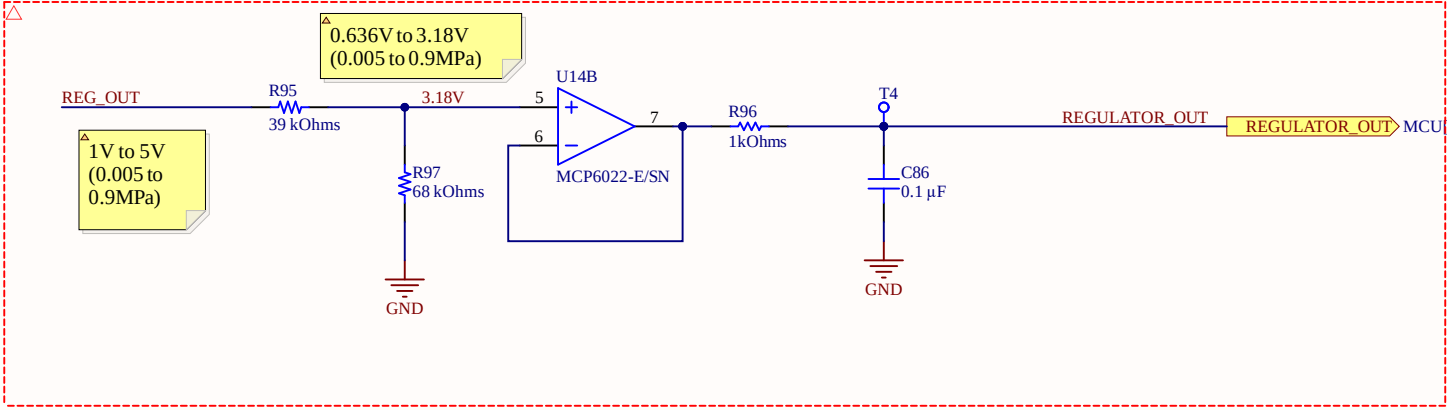
Regulator IN



Pin Out:
Brown = Vdc (24V)
Blue = GND
White = Analog Signal IN, sink type (4mA to 20mA)
Black = Analog Signal Monitoring (1V to 5V)

Rango de presion (0.005 to 0.9 MPa)

Regulator OUT



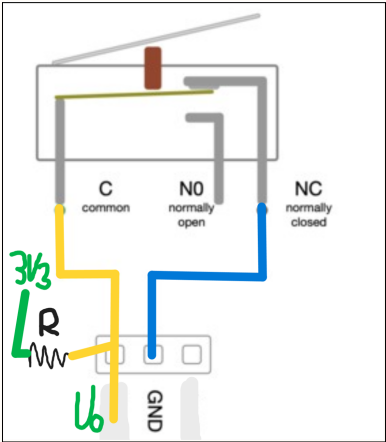
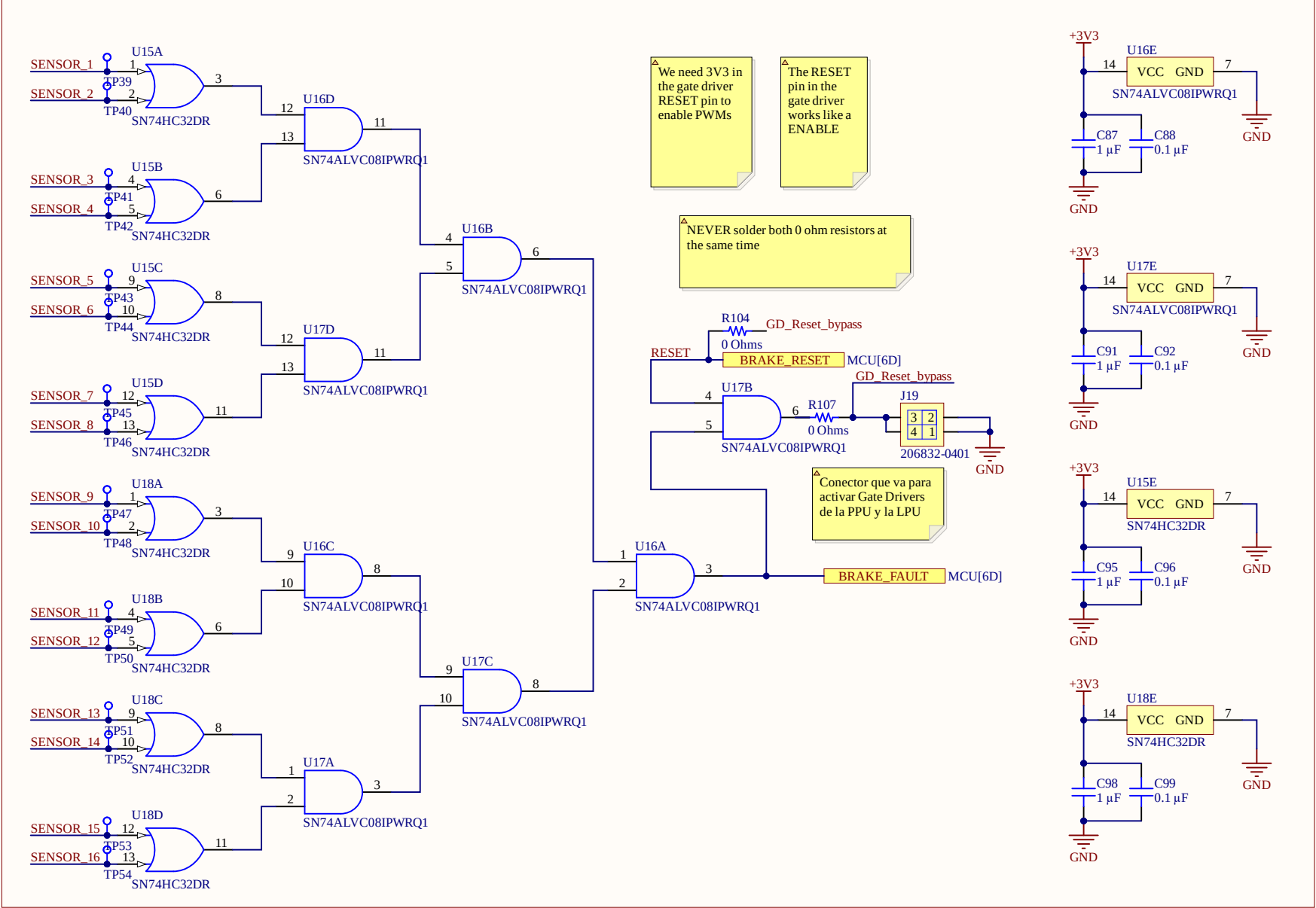
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HYPERLOOP UPV



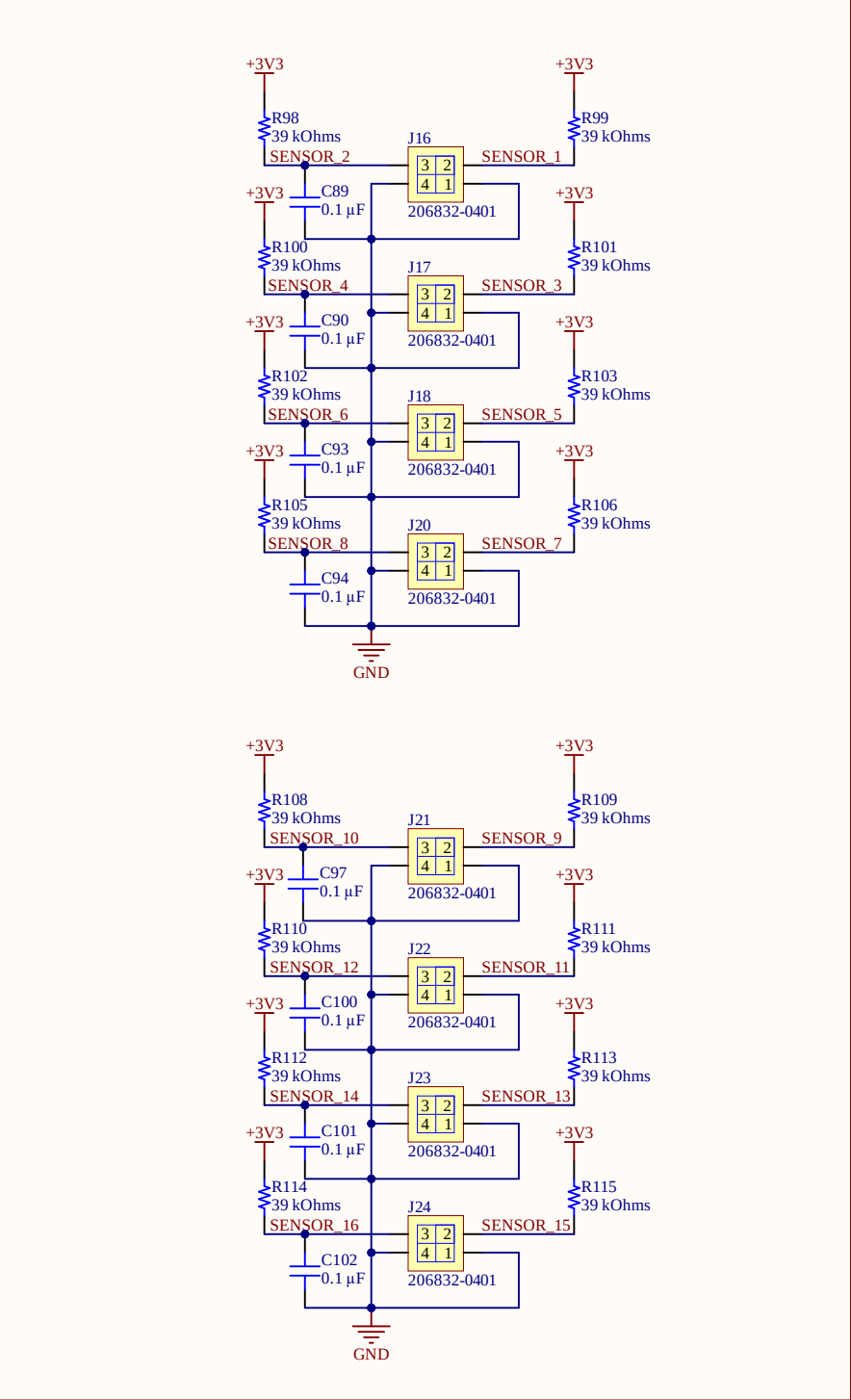
Limit switch logic



We should connect the NC pin of the limit switch sensor to GND and the NO pin floating

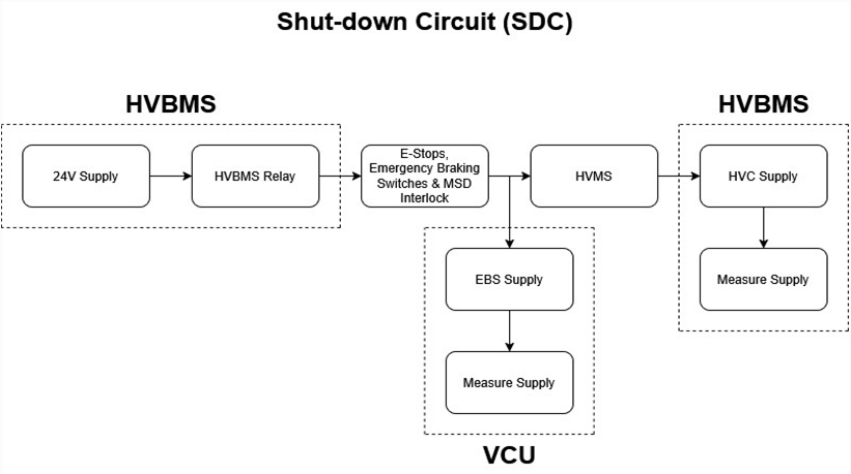
When the vehicle is braked, the limit switch is NOT pressed, it is pressed when the brakes are released

Limit switch connector

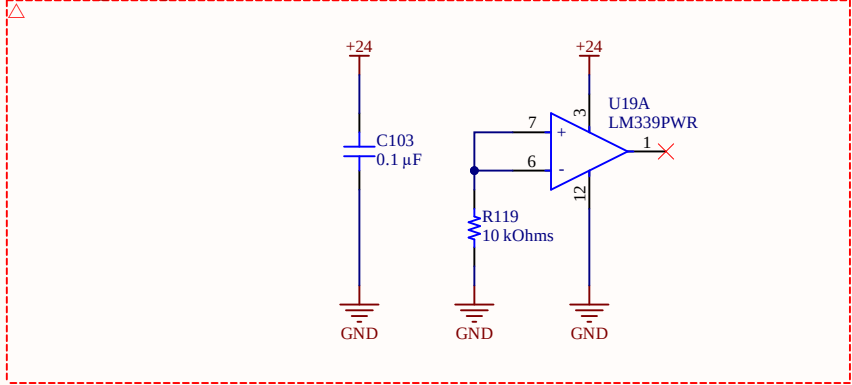


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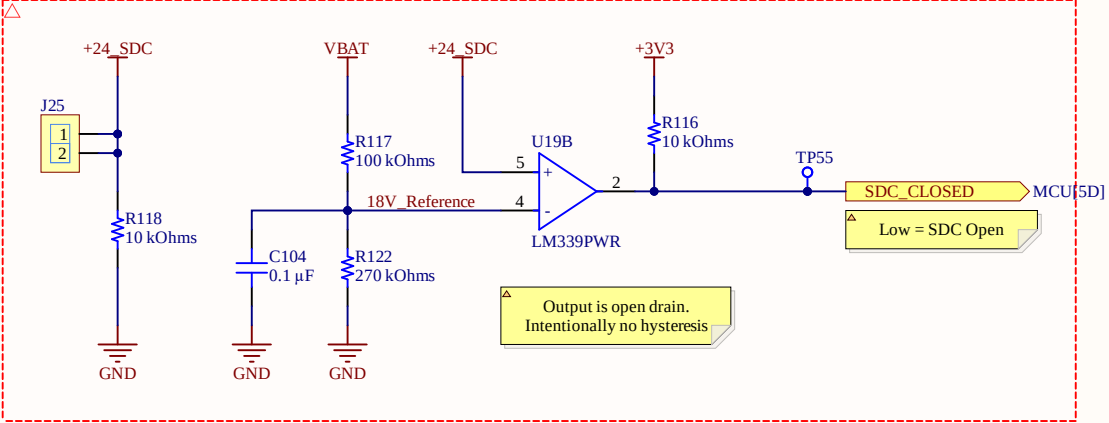




Decoupling & Unused Channels



SDC Input & Measurement



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